Your Name

Embedded Systems & ECE | IoT · Instrumentation & Control

Your City, Country | +1 555 000 0000 | you@example.com
linkedin.com/in/yourname | github.com/yourname | yourname.dev

Education

Example University May 2027 expected
Bachelor of Science in Electrical and Computer Engineering | GPA: 3.85/4.00 *City, Country*

- **Relevant Coursework:** Embedded Systems Design, Digital Signal Processing, Control Systems, Microcontroller Architecture, Real-Time Operating Systems (RTOS), PCB Layout.

Experience

Embedded Systems Intern Jun 2026 – Aug 2026
Example Telecom *City, Country*

- Developed bare-metal C firmware on STM32 for 5 Modbus RTU sensors over RS-485 at 115200 baud.
- Optimized SPI DMA transfers, reducing telemetry latency by 35% and preventing dropped packets.

Instrumentation & Control Intern Jan 2026 – May 2026
Example Dynamics *City, Country*

- Programmed an ESP32 PID loop regulating chamber temperature within ±0.5 °C across 20–80 °C.
- Integrated 4–20 mA current-loop transmitters and calibrated RTDs, improving accuracy by 15%.

IoT Firmware Intern May 2025 – Aug 2025
Acme Cloud Technologies *City, Country*

- Implemented deep-sleep modes for nRF52 BLE beacons, extending battery life from 8 to 22 months.
- Debugged hardware timing faults using logic analyzers and oscilloscopes to resolve intermittent I2C bus lockups.

Projects

Autonomous Balancing Robot | *C++, FreeRTOS, STM32* | GitHub Sep – Dec 2025

- Architected a FreeRTOS app with 4 tasks to process MPU-6050 IMU telemetry via I2C at 200 Hz.
- Implemented a PID controller stabilizing dual BLDC motors against tilt impulses up to 15 degrees.

LoRaWAN Industrial Sensor Node | *C, Zephyr RTOS, LoRaWAN* | GitHub Feb – May 2025

- Engineered Zephyr RTOS firmware transmitting pressure and vibration data over LoRaWAN up to 3 km.
- Configured sleep states, achieving an 18 uA quiescent current draw during telemetry intervals.

Skills

Languages & Systems: Embedded C, C++, Python, Assembly (ARM Cortex-M), FreeRTOS, Zephyr RTOS, Linux
Protocols & Busses: UART, USART, SPI, I2C, CAN bus, Modbus RTU, RS-485, BLE, LoRaWAN, MQTT
Hardware & Tools: KiCad, Altium, Multilayer PCB Design, Oscilloscopes, Logic Analyzers, JTAG/SWD, Soldering
Instrumentation & Control: PID Control, Feedback Systems, 4–20 mA Loops, MATLAB/Simulink

Certifications

Certified LabVIEW Associate Developer (CLAD) | IPC-A-610 Electronics Assembly and Soldering Certified